

TIC 230163860

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.667529 d
Epoch (t0) 2797.6497 BTJD
Duration 0.0400 d
Depth 3.966 % (39,660 ppm)
SNR 29.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

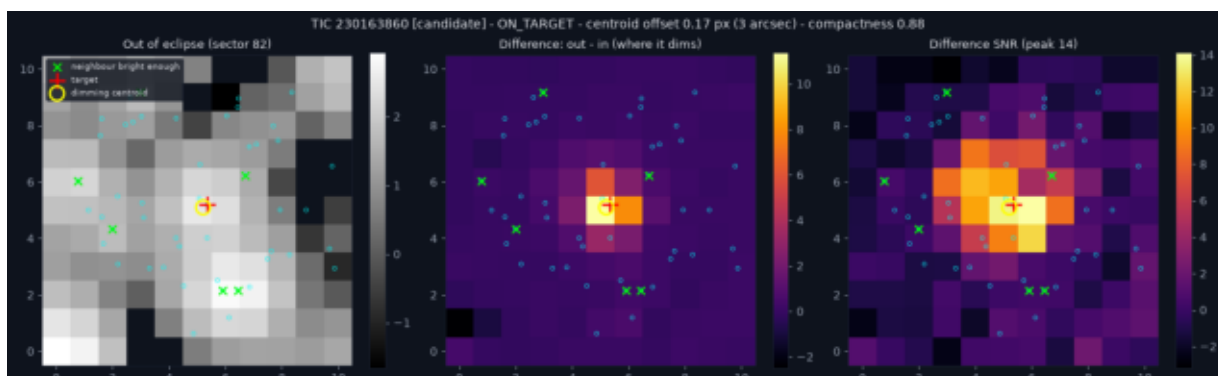
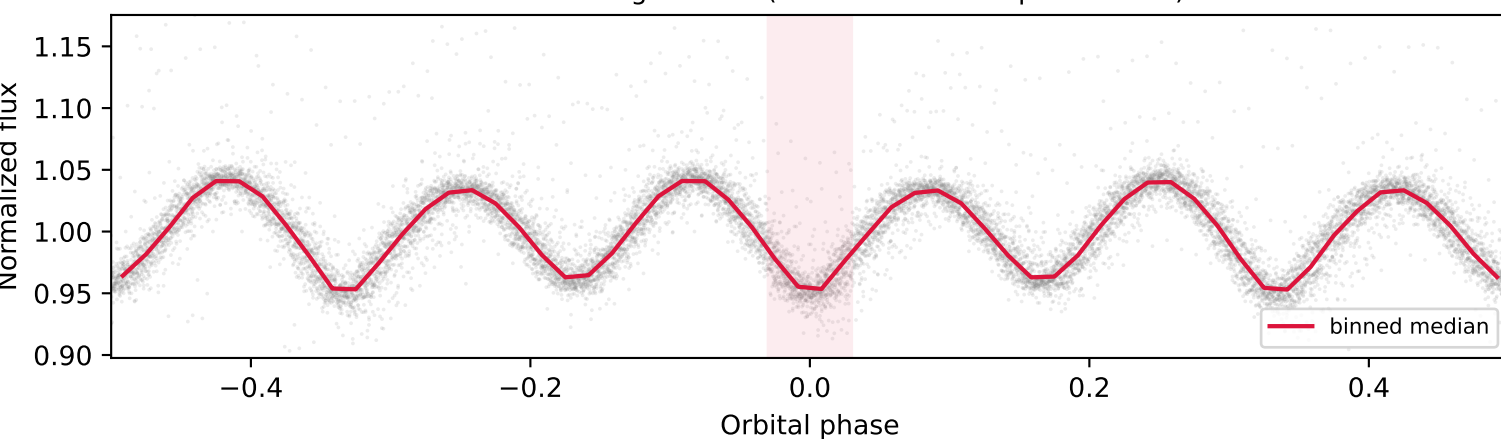
Eclipses observed 76 (gate ≥ 3)
Robust depth ratio 1.15 (gate ≥ 0.50)
Robust depth sigma 28.3 (gate ≥ 7.0)
Odd/even z-score 0.19 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 82 light-curve dip in this sector: 13.7 sigma t0 refined by -0.3375 phase
Difference-image peak SNR 14.14 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.17 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 230163855
1.89 px away, dTmag +3.14 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.